

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk: **Benzine (Petroleum Naphtha)**  
 Product Description: **Benzine (Petroleum Naphtha)**  
 Cat No. : B264-20; XXB264208LI; NC3522089  
 Synonyms Naphtha; Naphtha solvent; Petroleum distillates  
 CAS No 64742-89-8  
 Molecular Formula C5 to C10 hydrocarbons

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

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**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                    |
|------------------------------------------------------|--------------------|
| Flammable liquids                                    | Category 2 (H225)  |
| Aspiration Toxicity                                  | Category 1 (H304)  |
| Skin Corrosion/Irritation                            | Category 2 (H315)  |
| Reproductive Toxicity                                | Category 2 (H361f) |
| Specific target organ toxicity - (single exposure)   | Category 3 (H336)  |
| Specific target organ toxicity - (repeated exposure) | Category 2 (H373)  |
| Chronic aquatic toxicity                             | Category 2 (H411)  |

**Label Elements**


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## Signal Word

## Danger

### Hazard Statements

H225 - Highly flammable liquid and vapor  
H336 - May cause drowsiness or dizziness  
H304 - May be fatal if swallowed and enters airways  
H315 - Causes skin irritation  
H361f - Suspected of damaging fertility  
H373 - May cause damage to organs through prolonged or repeated exposure  
H411 - Toxic to aquatic life with long lasting effects

### Precautionary Statements

#### Prevention

P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P201 - Obtain special instructions before use  
P240 - Ground and bond container and receiving equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear eye protection/ face protection  
P280 - Wear protective gloves

#### Response

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P331 - Do NOT induce vomiting  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

#### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                    | CAS No     | Weight % |
|----------------------------------------------|------------|----------|
| Solvent naphtha (petroleum), light aliphatic | 64742-89-8 | 100      |
| Octane                                       | 111-65-9   | 1.5      |
| n-Heptane                                    | 142-82-5   | 1.16     |
| Xylenes (o-, m-, p- isomers)                 | 1330-20-7  | 0.13     |
| m-Xylene                                     | 108-38-3   | 0.05     |
| Ethylbenzene                                 | 100-41-4   | 0.03     |
| Benzene                                      | 71-43-2    | 0.015    |
| Toluene                                      | 108-88-3   | 0.0118   |

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## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### **General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### **Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### **Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### **Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.

#### **Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. Risk of serious damage to the lungs (by aspiration).

#### **Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

### Indication of any immediate medical attention and special treatment needed

#### **Notes to Physician**

Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

Do not use a solid water stream as it may scatter and spread fire.

### Special hazards arising from the substance or mixture

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Vapors may form explosive mixtures with air.

#### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrocarbons, Aldehydes.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

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## Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

## Environmental precautions

Do not flush into surface water or sanitary sewer system.

## Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Take precautionary measures against static discharges. Use spark-proof tools and explosion-proof equipment.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Flammables area.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                    | Malaysia | ACGIH TLV                     | OSHA PEL                                                                                                                                                                            |
|------------------------------|----------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Octane                       |          | TWA: 300 ppm                  | (Vacated) TWA: 300 ppm<br>(Vacated) TWA: 1450 mg/m <sup>3</sup><br>(Vacated) STEL: 375 ppm<br>(Vacated) STEL: 1800 mg/m <sup>3</sup><br>TWA: 500 ppm<br>TWA: 2350 mg/m <sup>3</sup> |
| n-Heptane                    |          | TWA: 400 ppm<br>STEL: 500 ppm | (Vacated) TWA: 400 ppm<br>(Vacated) TWA: 1600 mg/m <sup>3</sup><br>(Vacated) STEL: 500 ppm<br>(Vacated) STEL: 2000 mg/m <sup>3</sup><br>TWA: 500 ppm<br>TWA: 2000 mg/m <sup>3</sup> |
| Xylenes (o-, m-, p- isomers) |          | TWA: 20 ppm                   | (Vacated) TWA: 100 ppm<br>(Vacated) TWA: 435 mg/m <sup>3</sup><br>(Vacated) STEL: 150 ppm<br>(Vacated) STEL: 655 mg/m <sup>3</sup><br>TWA: 100 ppm<br>TWA: 435 mg/m <sup>3</sup>    |
| m-Xylene                     |          | TWA: 20 ppm                   |                                                                                                                                                                                     |
| Ethylbenzene                 |          | TWA: 20 ppm                   | (Vacated) TWA: 100 ppm<br>(Vacated) TWA: 435 mg/m <sup>3</sup>                                                                                                                      |

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|         |  |                                       |                                                                                                                                                                        |
|---------|--|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |  |                                       | (Vacated) STEL: 125 ppm<br>(Vacated) STEL: 545 mg/m <sup>3</sup><br>TWA: 100 ppm<br>TWA: 435 mg/m <sup>3</sup>                                                         |
| Benzene |  | TWA: 0.5 ppm<br>STEL: 2.5 ppm<br>Skin | (Vacated) TWA: 10 ppm<br>Ceiling: 25 ppm<br>(Vacated) STEL: 50 ppm<br>(Vacated) Ceiling: 25 ppm<br>TWA: 10 ppm<br>TWA: 1 ppm<br>STEL: 5 ppm                            |
| Toluene |  | TWA: 20 ppm                           | (Vacated) TWA: 100 ppm<br>(Vacated) TWA: 375 mg/m <sup>3</sup><br>Ceiling: 300 ppm<br>(Vacated) STEL: 150 ppm<br>(Vacated) STEL: 560 mg/m <sup>3</sup><br>TWA: 200 ppm |

| Component                    | European Union                                                                                                               | The United Kingdom                                                                                                         | Germany                                                                                                                                                                                                                                                                                                                            |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Octane                       |                                                                                                                              |                                                                                                                            | TWA: 500 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 2400 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 500 ppm (8 Stunden). MAK except Trimethylpentane isomers<br>TWA: 2400 mg/m <sup>3</sup> (8 Stunden). MAK except Trimethylpentane isomers<br>Höhepunkt: 1000 ppm<br>Höhepunkt: 4800 mg/m <sup>3</sup> |
| n-Heptane                    | TWA: 500 ppm (8h)<br>TWA: 2085 mg/m <sup>3</sup> (8h)                                                                        | STEL: 1500 ppm 15 min<br>STEL: 6255 mg/m <sup>3</sup> 15 min<br>TWA: 500 ppm 8 hr<br>TWA: 2085 mg/m <sup>3</sup> 8 hr      | TWA: 500 ppm (8 Stunden). AGW - exposure factor 1<br>TWA: 2100 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 1<br>TWA: 500 ppm (8 Stunden). MAK<br>TWA: 2100 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 500 ppm<br>Höhepunkt: 2100 mg/m <sup>3</sup>                                                                  |
| Xylenes (o-, m-, p- isomers) | TWA: 50 ppm (8h)<br>TWA: 221 mg/m <sup>3</sup> (8h)<br>STEL: 100 ppm (15min)<br>STEL: 442 mg/m <sup>3</sup> (15min)<br>Skin  | STEL: 100 ppm 15 min<br>STEL: 441 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 220 mg/m <sup>3</sup> 8 hr<br>Skin  | TWA: 50 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 220 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 50 ppm (8 Stunden). MAK all isomers<br>TWA: 220 mg/m <sup>3</sup> (8 Stunden). MAK all isomers<br>Höhepunkt: 100 ppm<br>Höhepunkt: 440 mg/m <sup>3</sup><br>Haut<br>Haut all isomers                   |
| m-Xylene                     | TWA: 50 ppm (8h)<br>TWA: 221 mg/m <sup>3</sup> (8h)<br>STEL: 100 ppm (15min)<br>STEL: 442 mg/m <sup>3</sup> (15min)<br>Skin  | STEL: 100 ppm 15 min<br>STEL: 441 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 220 mg/m <sup>3</sup> 8 hr<br>Skin  | TWA: 100 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 440 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 50 ppm (8 Stunden). MAK all isomers<br>TWA: 220 mg/m <sup>3</sup> (8 Stunden). MAK all isomers<br>Höhepunkt: 100 ppm<br>Höhepunkt: 440 mg/m <sup>3</sup><br>Haut<br>Haut all isomers                  |
| Ethylbenzene                 | TWA: 100 ppm (8h)<br>TWA: 442 mg/m <sup>3</sup> (8h)<br>STEL: 200 ppm (15min)<br>STEL: 884 mg/m <sup>3</sup> (15min)<br>Skin | STEL: 125 ppm 15 min<br>STEL: 552 mg/m <sup>3</sup> 15 min<br>TWA: 100 ppm 8 hr<br>TWA: 441 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 20 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 88 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 20 ppm (8 Stunden). MAK                                                                                                                                                                                 |

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|         |                                                                                                                                                                               |                                                                                                                                   |                                                                                                                                                                                                                                                                            |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |                                                                                                                                                                               |                                                                                                                                   | TWA: 88 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 40 ppm<br>Höhepunkt: 176 mg/m <sup>3</sup><br>Haut                                                                                                                                                                |
| Benzene | TWA: 0.2 ppm (8h)<br>TWA: 0.5 ppm (8h)<br>TWA: 1 ppm (8h)<br>TWA: 0.66 mg/m <sup>3</sup> (8h)<br>TWA: 1.65 mg/m <sup>3</sup> (8h)<br>TWA: 3.25 mg/m <sup>3</sup> (8h)<br>Skin | STEL: 3 ppm 15 min<br>STEL: 9.75 mg/m <sup>3</sup> 15 min<br>TWA: 1 ppm 8 hr<br>TWA: 3.25 mg/m <sup>3</sup> 8 hr<br>Carc.<br>Skin | Haut                                                                                                                                                                                                                                                                       |
| Toluene | TWA: 50 ppm (8hr)<br>TWA: 192 mg/m <sup>3</sup> (8hr)<br>STEL: 100 ppm (15min)<br>STEL: 384 mg/m <sup>3</sup> (15min)<br>Skin                                                 | STEL: 100 ppm 15 min<br>STEL: 384 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 191 mg/m <sup>3</sup> 8 hr<br>Skin         | TWA: 50 ppm (8 Stunden). AGW -<br>exposure factor 2<br>TWA: 190 mg/m <sup>3</sup> (8 Stunden). AGW<br>- exposure factor 2<br>TWA: 50 ppm (8 Stunden). MAK<br>TWA: 190 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 100 ppm<br>Höhepunkt: 380 mg/m <sup>3</sup><br>Haut |

## Exposure Controls

### Engineering Measures

Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

## Personal protective equipment

### Eye Protection

Wear safety glasses with side shields (or goggles)

### Hand Protection

Protective gloves

### Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

### Recommended Filter type:

Organic gases and vapours filter Type A Brown conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

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|                                                |                                 |                                             |
|------------------------------------------------|---------------------------------|---------------------------------------------|
| <b>Appearance</b>                              | Colorless                       |                                             |
| <b>Physical State</b>                          | Liquid                          |                                             |
| <b>Odor</b>                                    | Petroleum distillates           |                                             |
| <b>Odor Threshold</b>                          | No data available               |                                             |
| <b>pH</b>                                      | No information available        |                                             |
| <b>Melting Point/Range</b>                     | No data available               |                                             |
| <b>Softening Point</b>                         | No data available               |                                             |
| <b>Boiling Point/Range</b>                     | 118.5 - 140 °C / 245.3 - 284 °F |                                             |
| <b>Flash Point</b>                             | 14 - 18 °C / 57.2 - 64.4 °F     | <b>Method</b> - No information available    |
| <b>Evaporation Rate</b>                        | 1.3 (Butyl Acetate = 1.0)       |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable                  | Liquid                                      |
| <b>Explosion Limits</b>                        | No data available               |                                             |
| <b>Vapor Pressure</b>                          | 15 mmHg @ 20 °C                 |                                             |
| <b>Vapor Density</b>                           | 4.1 (Air = 1.0)                 | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.740                           |                                             |
| <b>Bulk Density</b>                            | Not applicable                  | Liquid                                      |
| <b>Water Solubility</b>                        | Insoluble                       |                                             |
| <b>Solubility in other solvents</b>            | No information available        |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                                 |                                             |
| <b>Component</b>                               | <b>log Pow</b>                  |                                             |
| Octane                                         | 5.18                            |                                             |
| n-Heptane                                      | 4.66                            |                                             |
| Xylenes (o-, m-, p- isomers)                   | 3.15                            |                                             |
| m-Xylene                                       | 3.2                             |                                             |
| Ethylbenzene                                   | 3.6                             |                                             |
| Benzene                                        | 2.13                            |                                             |
| Toluene                                        | 2.73                            |                                             |
| <b>Autoignition Temperature</b>                | 232 °C / 449.6 °F               |                                             |
| <b>Decomposition Temperature</b>               | No data available               |                                             |
| <b>Viscosity</b>                               | No data available               |                                             |
| <b>Explosive Properties</b>                    |                                 | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available        |                                             |
| <b>Molecular Formula</b>                       | C5 to C10 hydrocarbons          |                                             |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

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## Hazardous Polymerization Hazardous Reactions

No information available.  
None under normal processing.

## Conditions to Avoid

Incompatible products. Heat, flames and sparks. Keep away from open flames, hot surfaces and sources of ignition.

## Incompatible Materials

Strong oxidizing agents.

## Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrocarbons. Aldehydes.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

No acute toxicity information is available for this product

#### (a) acute toxicity;

##### Oral

Based on available data, the classification criteria are not met

##### Dermal

Based on available data, the classification criteria are not met

##### Inhalation

Based on available data, the classification criteria are not met

| Component                                    | LD50 Oral                 | LD50 Dermal                   | LC50 Inhalation                              |
|----------------------------------------------|---------------------------|-------------------------------|----------------------------------------------|
| Solvent naphtha (petroleum), light aliphatic | -                         | LD50 = 3000 mg/kg ( Rabbit )  | -                                            |
| Octane                                       | >5 g/kg (Rat)             | >2 g/kg (Rabbit)              | LC50 > 24.88 mg/L ( Rat ) 4 h                |
| n-Heptane                                    | >2000 mg/kg (rat)         | LD50 = 3000 mg/kg ( Rabbit )  | LC50 > 73.5 mg/L ( Rat ) 4 h                 |
| Xylenes (o-, m-, p- isomers)                 | LD50 = 3500 mg/kg ( Rat ) | LD50 > 4350 mg/kg ( Rabbit )  | 29.08 mg/L [MOE Risk Assessment Vol.1, 2002] |
| m-Xylene                                     | LD50 = 5 g/kg ( Rat )     | LD50 = 12.18 g/kg ( Rabbit )  | LC50 = 27124 mg/m <sup>3</sup> ( Rat ) 4 h   |
| Ethylbenzene                                 | 3500 mg/kg ( Rat )        | 15400 mg/kg ( Rabbit )        | 17.2 mg/L ( Rat ) 4 h                        |
| Benzene                                      | LD50 = 810 mg/kg ( Rat )  | LD50 > 8200 mg/kg ( Rabbit )  | LC50 = 44.66 mg/L ( Rat ) 4 h                |
| Toluene                                      | > 5000 mg/kg ( Rat )      | LD50 = 12000 mg/kg ( Rabbit ) | 26700 ppm ( Rat ) 1 h                        |

#### (b) skin corrosion/irritation;

Category 2

#### (c) serious eye damage/irritation;

No data available

#### (d) respiratory or skin sensitization;

##### Respiratory

No data available

##### Skin

No data available

#### (e) germ cell mutagenicity;

No data available

May cause heritable genetic damage



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(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component                                    | EU                       | UK | Germany | IARC     |
|----------------------------------------------|--------------------------|----|---------|----------|
| Solvent naphtha (petroleum), light aliphatic | Carc Cat. 1B<br>(note P) |    |         |          |
| Ethylbenzene                                 |                          |    |         | Group 2B |
| Benzene                                      | Carc Cat. 1A             |    | Cat. 1  | Group 1  |

(g) reproductive toxicity; Category 2

(h) STOT-single exposure; Category 3

Results / Target organs Central nervous system (CNS).

(i) STOT-repeated exposure; Category 2

Target Organs Central nervous system (CNS).

(j) aspiration hazard; Category 1

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects The product contains following substances which are hazardous for the environment. Contains a substance which is: Very toxic to aquatic organisms.

| Component                                    | Freshwater Fish                                                                                                                                                                                                                                                                          | Water Flea                                                                        | Freshwater Algae                                            | Microtox                |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------|
| Solvent naphtha (petroleum), light aliphatic |                                                                                                                                                                                                                                                                                          |                                                                                   | EC50: = 4700 mg/L, 72h<br>(Pseudokirchneriella subcapitata) |                         |
| Octane                                       |                                                                                                                                                                                                                                                                                          | EC50: = 0.38 mg/L, 48h<br>(water flea)                                            |                                                             | EC50 = 890 mg/L 30 min  |
| n-Heptane                                    | LC50: = 375.0 mg/L, 96h (Cichlid fish)                                                                                                                                                                                                                                                   | EC50: >10 mg/L/24h                                                                |                                                             |                         |
| Xylenes (o-, m-, p- isomers)                 | LC50: 30.26 - 40.75 mg/L, 96h static (Poecilia reticulata)<br>LC50: = 780 mg/L, 96h semi-static (Cyprinus carpio)<br>LC50: 23.53 - 29.97 mg/L, 96h static (Pimephales promelas)<br>LC50: > 780 mg/L, 96h (Cyprinus carpio)<br>LC50: 7.711 - 9.591 mg/L, 96h static (Lepomis macrochirus) | LC50: = 0.6 mg/L, 48h (Gammarus lacustris)<br>EC50: = 3.82 mg/L, 48h (water flea) |                                                             | EC50 = 0.0084 mg/L 24 h |

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|              |                                                                                                                                                                                                                                                                                                                                                                        |                                                                                              |                                                                                                                                                                                                                                                                 |                                                |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
|              | LC50: = 19 mg/L, 96h (Lepomis macrochirus)<br>LC50: 13.1 - 16.5 mg/L, 96h flow-through (Lepomis macrochirus)<br>LC50: 13.5 - 17.3 mg/L, 96h (Oncorhynchus mykiss)<br>LC50: 2.661 - 4.093 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: = 13.4 mg/L, 96h flow-through (Pimephales promelas)                                                                           |                                                                                              |                                                                                                                                                                                                                                                                 |                                                |
| m-Xylene     | LC50: = 12.9 mg/L, 96h semi-static (Poecilia reticulata)<br>LC50: 14.3 - 18 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 8.4 mg/L, 96h semi-static (Oncorhynchus mykiss)                                                                                                                                                                                    | EC50: 2.81 - 5.0 mg/L, 48h Static (Daphnia magna)                                            | EC50: = 4.9 mg/L, 72h static (Pseudokirchneriella subcapitata)                                                                                                                                                                                                  | EC50 = 0.0084 mg/L 24 h                        |
| Ethylbenzene | LC50: 9.1 - 15.6 mg/L, 96h static (Pimephales promelas)<br>LC50: 11.0 - 18.0 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: = 4.2 mg/L, 96h semi-static (Oncorhynchus mykiss)<br>LC50: 7.55 - 11 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 32 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 9.6 mg/L, 96h static (Poecilia reticulata)              | EC50: 1.8 - 2.4 mg/L, 48h (Daphnia magna)                                                    | EC50: 2.6 - 11.3 mg/L, 72h static (Pseudokirchneriella subcapitata)<br>EC50: 1.7 - 7.6 mg/L, 96h static (Pseudokirchneriella subcapitata)<br>EC50: > 438 mg/L, 96h (Pseudokirchneriella subcapitata)<br>EC50: = 4.6 mg/L, 72h (Pseudokirchneriella subcapitata) | EC50 = 9.68 mg/L 30 min<br>EC50 = 96 mg/L 24 h |
| Benzene      | LC50: = 22.49 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 5.3 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 70000 - 142000 µg/L, 96h static (Lepomis macrochirus)<br>LC50: = 28.6 mg/L, 96h static (Poecilia reticulata)<br>LC50: 22330 - 41160 µg/L, 96h static (Pimephales promelas)<br>LC50: 10.7 - 14.7 mg/L, 96h flow-through (Pimephales promelas) | EC50: = 10 mg/L, 48h (Daphnia magna)<br>EC50: 8.76 - 15.6 mg/L, 48h Static (Daphnia magna)   | EC50: = 29 mg/L, 72h (Pseudokirchneriella subcapitata)                                                                                                                                                                                                          |                                                |
| Toluene      | 50-70 mg/L LC50 96 h<br>5-7 mg/L LC50 96 h<br>15-19 mg/L LC50 96 h<br>28 mg/L LC50 96 h<br>12 mg/L LC50 96 h                                                                                                                                                                                                                                                           | EC50: = 11.5 mg/L, 48h (Daphnia magna)<br>EC50: 5.46 - 9.83 mg/L, 48h Static (Daphnia magna) | EC50: = 12.5 mg/L, 72h static (Pseudokirchneriella subcapitata)<br>EC50: > 433 mg/L, 96h (Pseudokirchneriella subcapitata)                                                                                                                                      | EC50 = 19.7 mg/L 30 min                        |

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Benzine (Petroleum Naphtha)

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|  |  |  |  |  |
|--|--|--|--|--|
|  |  |  |  |  |
|--|--|--|--|--|

## Persistence and degradability

**Persistence** Insoluble in water, Persistence is unlikely, based on information available.

| Component                      | Degradability |
|--------------------------------|---------------|
| Toluene<br>108-88-3 ( 0.0118 ) | 86% (20d)     |

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## Bioaccumulative potential

May have some potential to bioaccumulate

| Component                    | log Pow | Bioconcentration factor (BCF) |
|------------------------------|---------|-------------------------------|
| Octane                       | 5.18    | No data available             |
| n-Heptane                    | 4.66    | No data available             |
| Xylenes (o-, m-, p- isomers) | 3.15    | 0.6 - 15 dimensionless        |
| m-Xylene                     | 3.2     | No data available             |
| Ethylbenzene                 | 3.6     | 15 dimensionless              |
| Benzene                      | 2.13    | 3.5 - 4.4 dimensionless       |
| Toluene                      | 2.73    | 90                            |

## Mobility in soil

Spillage unlikely to penetrate soil. The product is insoluble and floats on water. The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. . Is not likely mobile in the environment due its low water solubility. Will likely be mobile in the environment due to its volatility.

## Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

### **Other Information**

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Can be landfilled or incinerated, when in compliance with local regulations Do not let this chemical enter the environment Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

|                      |                               |
|----------------------|-------------------------------|
| UN-No                | UN1268                        |
| Hazard Class         | 3                             |
| Packing Group        | II                            |
| Proper Shipping Name | Petroleum distillates, n.o.s. |

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## Road and Rail Transport

UN-No UN1268  
Hazard Class 3  
Packing Group III  
Proper Shipping Name Petroleum distillates, n.o.s.

## IATA

UN-No UN1268  
Hazard Class 3  
Packing Group II  
Proper Shipping Name Petroleum distillates, n.o.s.

Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component                                    | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|----------------------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Solvent naphtha (petroleum), light aliphatic | 265-192-2 | X    | X   | X     | -    |      | X     | X    | KE-31661 |
| Octane                                       | 203-892-1 | X    | X   | X     | X    | X    | X     | X    | KE-26612 |
| n-Heptane                                    | 205-563-8 | X    | X   | X     | X    | X    | X     | X    | KE-18271 |
| Xylenes (o-, m-, p- isomers)                 | 215-535-7 | X    | X   | X     | X    | X    | X     | X    | KE-35427 |
| m-Xylene                                     | 203-576-3 | X    | X   | X     | X    | X    | X     | X    | KE-35428 |
| Ethylbenzene                                 | 202-849-4 | X    | X   | X     | X    | X    | X     | X    | KE-13532 |
| Benzene                                      | 200-753-7 | X    | X   | X     | X    | X    | X     | X    | KE-02150 |
| Toluene                                      | 203-625-9 | X    | X   | X     | X    | X    | X     | X    | KE-33936 |

| Component                    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Xylenes (o-, m-, p- isomers) |                                                                                           |                                                                                          |                            | Annex I - Y42                      |
| Toluene                      |                                                                                           |                                                                                          |                            | Annex I - Y42                      |

### National Regulations

Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

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**WEL** - Workplace Exposure Limit  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**RPE** - Respiratory Protective Equipment  
**LC50** - Lethal Concentration 50%  
**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date** 24-Mar-2025  
**Revision Summary** Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**